ships and LNG carriers.

Ships falling into the scope of  EEDI requirement can use their attained EEDI calculated in accordance with the 2018 Guidelines on the method of calculation of the attained EEDI for new ships (resolution MEPC.308(73), as amended, the "EEDI Calculation Guidelines" hereafter) as the attained EEXI if the value of the attained EEDI is equal to or less than that of the required EEXI.

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SOURCE: MEPC.333(76).pdf
new ships (resolution MEPC.308(73), as amended, the "EEDI Calculation Guidelines" hereafter) as the attained EEXI if the value of the attained EEDI is equal to or less than that of the required EEXI.

2.2 Parameters

For calculation of the attained EEXI by the formula in paragraph 2.1, parameters under the EEDI Calculation Guidelines apply, unless expressly provided otherwise. In referring to the aforementioned guidelines, the terminology "EEDI" should be read as "EEXI".

2.2.1 PME(i) ; Power of main engines

In cases where overridable Shaft / Engine Power Limitation is installed in accordance with the 2021 Guidelines on the shaft / engine power limit to comply with the EEXI requirements and use of a power reserve (resolution MEPC.335(76)), PME(i) is 83% of the limited installed power (MCRlim) or 75% of the original installed power (MCR), whichever is lower, for each main engine (i). In cases where the overridable Shaft / Engine Power Limitation and shaft generator(s) are installed, in referring to paragraph 2.2.5.2 (option 1) of the EEDI Calculation Guidelines, "MCRME" should be read as "MCRlim".
SOURCE: MEPC.333(76).pdf
engine (i). In cases where the overridable Shaft / Engine Power Limitation and shaft generator(s) are installed, in referring to paragraph 2.2.5.2 (option 1) of the EEDI Calculation Guidelines, "MCRME" should be read as "MCRlim".

For LNG carriers having steam turbine or diesel electric propulsion, PME(i) is 83% of the limited installed power (MCRlim, MPPlim), divided by the electrical efficiency in case of diesel electric propulsion system, for each main engine (i). For LNG carriers, the power from combustion of
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the excessive natural boil-off gas in the engines or boilers to avoid releasing to the atmosphere

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SOURCE: MEPC.333(76).pdf
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the excessive natural boil-off gas in the engines or boilers to avoid releasing to the atmosphere or unnecessary thermal oxidation should be deducted from PME(i) with the approval of the verifier.

2.2.2 PAE(i) ; Power of auxiliary engines

2.2.2.1 PAE(i) is calculated in accordance with paragraph 2.2.5.6 of the EEDI Calculation Guidelines.

2.2.2.2 For ships where power of auxiliary engines ( PAE) value calculated by paragraphs 2.2.5.6.1 to 2.2.5.6.3 of the EEDI Calculation Guidelines is significantly diff erent from the total power used at normal seagoing, e.g. in cases of passenger ships, the PAE value should be estimated by the consumed electric power (excluding propulsion) in conditions when the ship is engaged in a voyage at reference speed ( Vref) as given in the electric power table, divided by the average efficiency of the generator(s) weighted by power (see appendix 2 of the EEDI Calculation Guidelines).

2.2.2.3 In cases where the electric power table is not available, the P AE value may be approximated either by:

.1 annual average figure of P AE at sea from onboard monitoring obtained prior

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SOURCE: MEPC.333(76).pdf
EEDI Calculation Guidelines).

2.2.2.3 In cases where the electric power table is not available, the P AE value may be approximated either by:

.1 annual average figure of P AE at sea from onboard monitoring obtained prior to the EEXI certification;

.2 for cruise passenger ships, approximated value of power of auxiliary engines (PAE,app), as defined below:

, = 0.1193 ×  + 1814.4  [kW]

.3 for ro-ro passenger ships, approximated value of power of auxiliary engines (PAE,app), as defined below:

, = 0.866 × 0.732  [kW]

2.2.3 Vref ; Ship speed

2.2.3.1 For ships falling into the scope of the EEDI requirement, the ship speed V ref should be obtained from an approved speed-power curve as defined in the 2014 Guidelines on survey and certification of the Energy Efficiency Design Index (EEDI) , as amended (resolution MEPC.254(67), as amended).

2.2.3.2 For ships not falling into the scope of the EEDI requirement, the ship speed Vref should be obtained from an estimated speed-power curve as defined in the 2021 Guidelines on survey and certification of the attained EEXI (resolution MEPC.334(76)).

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SOURCE: MEPC.333(76).pdf
be obtained from an estimated speed-power curve as defined in the 2021 Guidelines on survey and certification of the attained EEXI (resolution MEPC.334(76)).

2.2.3.3 For ships not falling into the scope of the EEDI requirement but whose sea trial results, which may have been calibrated by the tank test, under the EEDI draught and the sea condition as specified in paragraph 2.2.2 of the EEDI Calculation Guidelines are included in the sea trial report, the ship speed Vref may be obtained from the sea trial report:

 = ,  ×

,

1
3
  [knot]


where,

VS,EEDI, is the sea trial service speed under the EEDI draught; and

PS,EEDI is power of the main engine corresponding to VS,EEDI.

2.2.3.4 For containerships, bulk carriers or tankers not falling into the scope of the EEDI

requirement but whose sea trial results, which may have been calibrated by the tank test, under

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SOURCE: MEPC.333(76).pdf
2.2.3.4 For containerships, bulk carriers or tankers not falling into the scope of the EEDI requirement but whose sea trial results, which may have been calibrated by the tank test, under the design load draught and sea condition as specified in paragraph 2.2.2 of the EEDI Calculation Guidelines are included in the sea trial report, the ship speed Vref may be obtained from the sea trial report:

$$V_{ref} = \frac{1}{3} \times \left( \frac{2}{9} \times V_{S,service} \times \sqrt{\frac{1}{3}} \right) \text{ [knot]}$$

where,

$V_{S,service}$ is the sea trial service speed under the design load draught;